

MD KAWSER BEPARY

[\[Website\]](#) [\[LinkedIn\]](#) [\[GitHub\]](#) [\[Google Scholar\]](#)

Email : mkbepary.buet13@gmail.com

Mobile : +1-256-417-9343

Address: Gainesville, FL, USA

PROFESSIONAL SUMMARY

Hardware security and verification engineer specializing in secure SoC/ASIC design, pre-silicon security verification, and resilient semiconductor system development. Background includes side-channel vulnerability analysis, RTL and formal verification, embedded firmware for secure boot and cryptographic services, static RTL security checking, and FPGA-based reliability characterization. Experienced with C/C++, Python, SystemVerilog, FPGA prototyping, and EDA tools and flows, with a proven ability to translate advanced research into production-oriented security solutions across architecture, verification, and firmware teams.

EDUCATION

- **University of Florida** Gainesville, FL
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering Jan 2021 - Aug 2025
Advisor: Prof. Mark Tehranipoor CGPA: **3.95/4.00**
- **The University of Alabama in Huntsville** Huntsville, AL
Master of Science (M.Sc.) in Computer Engineering Aug 2019 - Dec 2020
Advisor: Prof. M. Tauhidur Rahman CGPA: **3.71/4.00**
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering Jul 2014 - Oct 2018
CGPA: **3.51/4.00**

EXPERIENCE

- **Caspia Technologies, Inc.** Gainesville, FL
Senior R&D Engineer Sep 2025 - Present
 - Designed and developed an end-to-end agentic security verification flow around CODAx, a static RTL security tool, to support pre-silicon verification of complex SoC designs.
 - Performed R&D and QA for CODAx, a static rule-based checker for RTL designs, refining security and fault-oriented rules, reducing noise, and improving coverage of critical design weaknesses.
 - Built an agentic post-analysis workflow for CODAx that enabled autonomous re-verification of violations, risk scoring, and RTL or architectural mitigation recommendations.
 - Developed automated RTL mitigation generation to address detected violations and support iterative repair and re-verification within the security analysis loop.
 - Designed an LLM-based flow to identify security-critical RTL assets and support formal property generation and security verification in SVx.
- **Synopsys, Inc.** Sunnyvale, CA
Graduate Technical Intern (Security IP Team), Solutions Group May 2023 - Dec 2023
 - Developed embedded applications for secure boot, TLS communication, and cryptographic services under DARPA-funded AISS (Automatic Implementation of Secure Silicon) project.
 - Implemented ASCON lightweight cryptographic protocols for embedded firmware security, providing authenticated encryption and hashing for resource-constrained, RTOS-based platforms.
 - Integrated ASCON authenticated encryption and hashing hardware module into the Synopsys security engine subsystem and implemented a masking scheme to improve side-channel resiliency.
- **FICS Research, Department of ECE, University of Florida** Gainesville, FL
Graduate Research Assistant Jan 2021 - Aug 2025
 - Built a Python-based framework to estimate IP/SoC power side-channel resiliency at RTL for a DARPA-funded AISS project, providing actionable metrics in collaboration with Synopsys.
 - Implemented an automated CAD toolflow to evaluate physical-design aware EM side-channel vulnerability at gate-level, enabling fast pre-silicon security sign-off.
 - Developed machine learning-based side-channel leakage modeling framework at the pre-silicon stage, enhancing the security of cryptographic designs against physical attacks.
 - Created an EM-based side-channel disassembler to reconstruct executed microcontroller instructions, enabling robust analysis in secure embedded systems.
 - Devised a PMU-based approach for detecting CPU workloads and mitigating cache side-channel vulnerabilities to counter speculative execution attacks.
- **Department of ECE, University of Florida** Gainesville, FL
Graduate Teaching Assistant - Intro to Hardware Security and Trust Aug 2022 - Dec 2022
 - Designed hands-on labs and tutorials using industry-standard EDA tools, guiding students through VLSI synthesis, functional verification, and hardware security analysis.

- Managed grading, projects, and exams while mentoring students on hardware security and verification methodologies.

- **Department of ECE, The University of Alabama in Huntsville**

Graduate Research Assistant

Huntsville, AL

Aug 2019 - Dec 2020

- Developed C++ based FPGA test infrastructure to interface with commercial DDR3 DRAM chips and execute configurable read/write experiments with tunable timing parameters.
- Conducted stress testing and soft-error analysis on DRAM modules to evaluate memory security vulnerabilities and reliability degradation under voltage, temperature, and aging stress conditions.

- **Department of ECE, The University of Alabama in Huntsville**

Graduate Teaching Assistant - Computer Organization, Intro to Computer Architecture

Huntsville, AL

Aug 2020 - Dec 2020

- Supported 100+ graduate/undergraduate students with grading, projects, and exams covering computer organization, architecture, memory hierarchy, assembly language, and MIPS ISA concepts.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Rust, Verilog, SystemVerilog, VHDL, Assembly (RISC-V/MIPS)
- **Verification & EDA Tools:**
 - **RTL Design & Synthesis:** Synopsys Design Compiler, Xilinx Vivado, Cadence Genus, ABC
 - **Functional Verification:** ModelSim, Synopsys VCS, SimVision, SystemVerilog Assertions (SVA), UVM
 - **Formal Verification:** Cadence JasperGold, Synopsys Formality
 - **Physical Design:** Cadence Innovus, Virtuoso, Synopsys ICC2
 - **Other EDA Tools:** Synopsys PrimeTime, TetraMax, StarRC
- **Scripting:** Tcl, Unix Shell, GNU Make, CMake
- **Data Analysis, ML & LLMs:** Scikit-learn, TensorFlow, Keras, PyTorch, LLM API integration (OpenAI/Anthropic), and agentic workflow design for automated RTL and EDA analysis
- **Other Software and Tools:** Linux, Git, PAPI, Perforce, LaTeX

SELECTED PUBLICATIONS

- HA Shaikh, **MK Bepary**, F Rahman, J Lee. "PFx: An Agentic Framework for Automated System-level Dynamic Security Verification." in Annual GOMACTech Workshop 2026, USA.
- **MK Bepary**, T Zhang, J Zhou, et al., "Towards Efficient Gate-Level Electromagnetic Side-Channel Leakage Modeling and Vulnerability Assessment," Journal of Hardware and Systems Security, 2025. ([link](#))
- **MK Bepary**, T Zhang, F Farahmandi and M Tehranipoor. "PreSCAN: A Comprehensive Review of Pre-Silicon Physical Side-Channel Vulnerability Assessment Methodologies." Chips 3, no. 3 (2024). ([link](#))
- T Rahman, **MK Bepary**, MSUI Haque, M Tehranipoor and F Rahman, "Design and Security-Mitigation of Custom and Configurable Hardware Cryptosystems," 2023 IEEE 16th DCAS, Denton, TX, USA, 2023, pp. 1-6. ([link](#))
- **MK Bepary**, BMSB Talukder, and MT Rahman. "DRAM Retention Behavior with Accelerated Aging in Commercial Chips." Applied Sciences 12, no. 9 (2022): 4332. ([link](#))
- B Ahmed, **MK Bepary**, N Pundir, M Borza, et al. "Quantifiable Assurance: From IPs to Platforms." Cryptology ePrint Archive (2021). ([link](#))

RELEVANT COURSEWORK

- **Computer Engineering:** Computer Architecture, System-on-Chip Design, Reconfigurable Computing, VLSI HDL/Modeling, Advanced Computer Architecture, Parallel Programming, Advanced VLSI
- **Hardware Security:** Cybersecurity Engineering, Hardware Security and Trust, CAD for Hardware Security Verification, Advanced Hardware Security
- **Machine Learning:** Data Mining, Intro to Machine Learning, Probability and Statistics

SELECTED ACADEMIC PROJECTS

- **System-on-Chip (SoC) Design:** Built a MicroBlaze-based SoC with AXI/APB peripheral integration and developed bare-metal C applications for on-chip communication and control (2023).
- **Convolution and DRAM DMA on FPGA:** Implemented a 1-D time-domain convolution accelerator and DRAM DMA interface on a ZedBoard FPGA to improve throughput using parallelism (2024).
- **Property-based Security Verification of an RSA Controller:** Used SystemVerilog Assertions in Cadence JasperGold to verify functional and security properties of an RSA hardware module (2022).
- **Implementation of Efficient AES Crypto Engine:** Developed an efficient VHDL implementation of AES 128-bit encryption/decryption engine using pipelined architecture on Xilinx Artix-7 FPGA board. (2020)

HONORS AND AWARDS

- 1st Place in 2022 IEEE HOST Microelectronics Security Challenge ([github](#)), Undergrad Dean's List (2015), Education Board Merit Scholarship (Secondary and Higher Secondary).